Every 0/1 polytope has Chvatal–Gomory rank zero

Conjecture 00000003973

zbw123456 18 October 2026

Use the standard meaning of a 0/1 polytope: $P = \operatorname{conv}(V)$ for $V \subseteq \{0,1\}^n$. Its integer hull is $P_I = \operatorname{conv}(P \cap \mathbb{Z}^n)$. Since every vertex in V is integral and P is convex,

$$P = \operatorname{conv}(V) \subseteq \operatorname{conv}(P \cap \mathbb{Z}^n) \subseteq P.$$

Therefore $P_I = P$.

The CG closure is the intersection of P with all cuts $a(x) \le \lfloor b \rfloor$, where a is a real linear functional integer-valued on $\mathbb{Z}^n$ and $a(y) \le b$ for all $y \in P$. These are exactly integer-coefficient CG cuts. The CG rank is the least nonnegative k for which $\mathrm{CG}^k(P) = P_I$. The zeroth iterate is P, so the equality already holds at $k = 0$. Hence every 0/1 polytope has rank zero.

In particular, no family of such polytopes, including partition polytopes in this class, can have ranks at least cn^2 for a fixed $c > 0$ and all sufficiently large n. Taking a positive n above the threshold would give $0 \ge cn^2 > 0$. The upper-bound clause alone is trivial, but the quadratic-order attainment and $\Theta(n^2)$ supremum clauses are false. In every positive dimension the class is nonempty (a singleton integer vertex suffices), and its set of ranks is $\{0\}$.

Scope. A polytope with fractional vertices merely contained in $[0,1]^n$ is a different class. The proof uses the standard vertex-based term *0/1 polytope* present in both language versions and makes no claim about arbitrary relaxations inside the cube.

Lean correspondence. Lean defines the integer lattice, real convex hull, integer hull, CG closure, and rank as an `IsLeast` statement. `integerHull_eq` explicitly constructs integer coordinates for 0/1 vertices and proves both inclusions. `rank_zero` and `no_positive_rank` prove the exact rank. `no_quadratic_family` rules out every positive eventual quadratic lower bound. No rank or integer-hull assertion is assumed.

Reproduction. The standalone project pins Lean 4.19.0 and Mathlib commit:

`c44e0c8ee63ca166450922a373c7409c5d26b00b`

Run `sh verify.sh`; the Lean source prints the main results' axiom dependencies. No proof placeholders, native evaluator bypass, or custom axioms are used.

Source. The official bilingual statement is included as `conjecture.md`, checked at repository commit

`cce2d90cb298de9595798eb674d2c80712100532`